Quantifiers

Specifying the number of times part of a pattern appears can be very useful. There are many cases where string data contains well-defined structure, which might be of interest in a program. Think along the lines of product serial numbers, ISBN's, for example. There is a general-purpose quantifier syntax, which is an open brace {, followed by a decimal integer (between *0* and *65535*), followed by a closing brace }. To define a range, add a comma after the first number, then an optional second integer (which must be larger than the first), and finally a closing brace }. For convenience, there are also some single character quantifiers: ? is "zero or one", + is "one or more", * is "zero or more". Put that all together and you have something like this:

Table 8-30. *PCRE quantifiers*

`{5,}`	At least five
`{0,1}`, `?`	Zero or one
`{1,}`, `+`	One or more
`{0,}`, `*`	Zero or more
`{42}`	Exactly forty-two

Quantifiers, be they single characters or in the general form, are placed immediately after the part of the expression to be quantified. For example:

Table 8-31. *PCRE quantifier examples*

`\d*`	Any number of digits, including none
`\w{7}\b`	Exactly seven "word" characters immediately followed by boundary
`\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}`	Four groups of up to three digits, separated by a literal dot: an IPv4 address.

Now, just to confuse things a little, the ? metacharacter has a double purpose for quantifying. If placed immediately after the closing brace of a general-purpose expression, it changes the *greediness* of the match (matches are greedy by default; the ? modifier flips this to non-greedy). A non-greedy match simply means the PCRE engine switches to matching the minimum portion of the sample to satisfy the pattern. The following examples demonstrate this: